



Submitted By:

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Assignment:

Lab No. 2

Submitted To:

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For the fulfillment of,

Object-Oriented Programming Lab

Department of Computer Engineering

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Task No. 1:

The Geometry Calculator (Operators & Math)

Objective: Use Math methods and arithmetic operators.

- Create a program that stores the radius of a circle as a double.
- Calculate the area ($\text{Area} = \pi \times r^2$). Use Math.PI and Math.Pow().
- Print the result. Then, use Type Casting to print the area as a whole integer.

Solution:

```
public class LAB_2{  
    public static void main(String[] args){  
        double radius=5.9d;    double  
        area=Math.PI*Math.pow(radius,2);  
        System.out.println("Area in double "+area);  
        int area_in_int=(int) area;  
        System.out.println("Area in int "+area_in_int);  
    }  
}
```

Output:

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\In  
Area in double 109.3588402714607  
Area in int 109  
  
Process finished with exit code 0
```

Task No. 2:

String Analysis Tool (Strings)

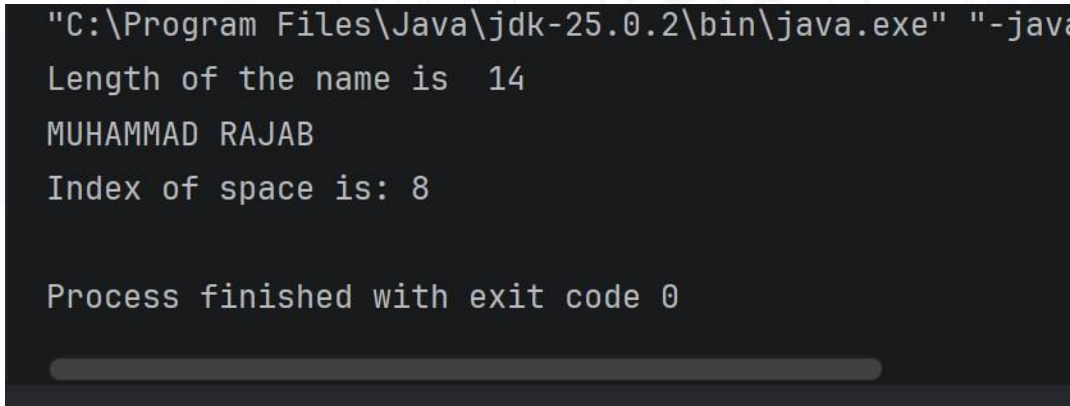
Objective: Manipulate and analyze text data.

- Create a String variable full Name containing your first and last name.
- Print the length of the string.
- Convert the name to all uppercase.
- Use index of () to find the position of the space between the names.

Solution:

```
public class LAB_2{    public
static void main(String[] args){    String
full_name="Muhammad Rajab";
    System.out.println( "Length of the name is "+full_name.length());
    System.out.println(full_name.toUpperCase());
    System.out.println("Index of space is: "+full_name.indexOf(" "));
} }
```

Output:



```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-java
Length of the name is 14
MUHAMMAD RAJAB
Index of space is: 8

Process finished with exit code 0
```

Task No. 3:

Automatic Grade Converter (Type Casting)

Objective: Narrowing Casting and Output.

- Declare a double variable percentage = 85.99.
- Manually cast this percentage into an int.
- Print: "Your raw score is: 85.99% but your rounded grade is: 85%".

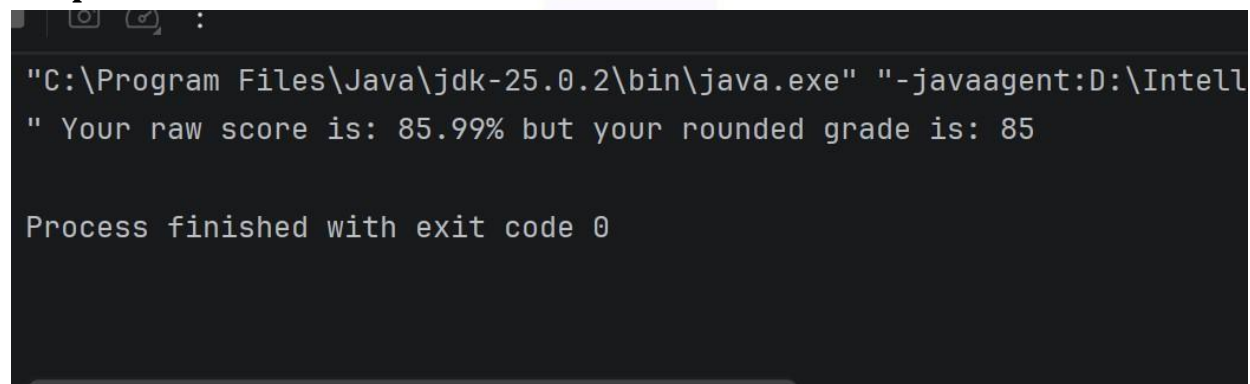
Solution:

```

public class LAB_2{
    public static void main(String[] args){
double percentage=85.99d;          int
per=(int)percentage;
        System.out.println("\n Your raw score is: "+percentage+"% but your rounded
grade is: "+per);
    }
}

```

Output:



```

"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\IntelliJ\lib\idea_rt.jar=12121:D:\IntelliJ\bin" -Dfile.encoding=UTF-8
" Your raw score is: 85.99% but your rounded grade is: 85

Process finished with exit code 0

```

Task No. 4:

The Logical Gatekeeper (Logical Operators)

Objective: Use &&, ||, and!

- Create two Boolean variables: Has ID and isOver18.
- Write an if statement that prints "Access Granted" only if both are true.
- Write another statement that prints "Special Guest" if they have an ID OR they are over 18.

Solution:

```

public class LAB_2{    public static
void main(String[] args){
    Scanner scan=new Scanner(System.in);
    System.out.println("Enter a boolean (hasID):");          boolean
hasID= scan.nextBoolean();
    System.out.println("Enter a boolean (is over18):");
    boolean isover18= scan.nextBoolean();
    if(hasID&&isover18){
        System.out.println("Access Granted");
    }
}

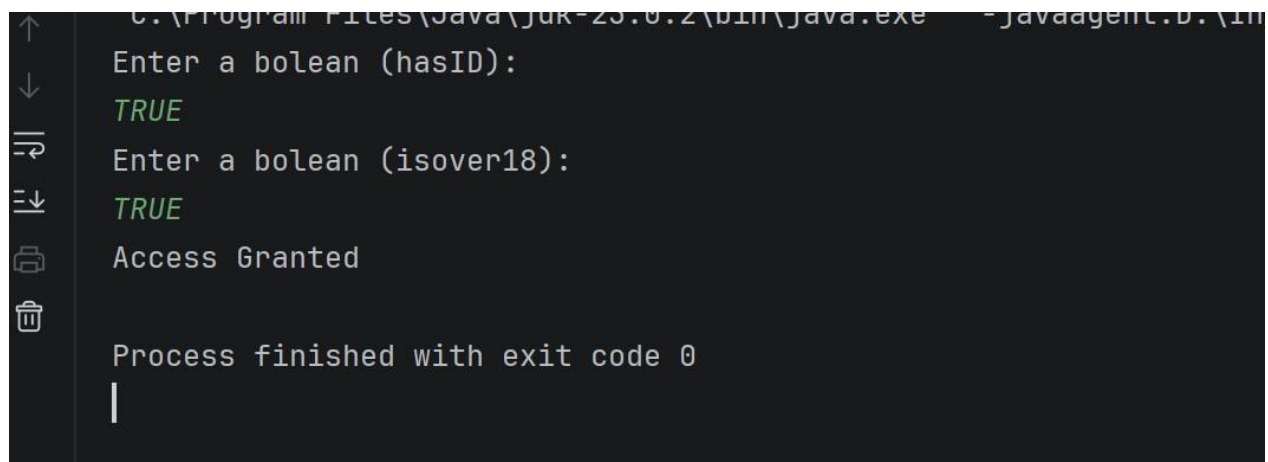
```

```

    } else if (hasID||isover18) {
        System.out.println("Special Guest");
    }
} }

```

Output:



```

C:\Program Files\Java\jdk-23.0.2\bin\java.exe -javaagent:C:\In
Enter a boolean (hasID):
TRUE
Enter a boolean (isover18):
TRUE
Access Granted
Process finished with exit code 0
|

```

Task No. 5:

Random Number Matcher (Math & Booleans)

Objective: Use Math. Random() and comparisons.

- Generate a random number between 1 and 10. Hint: Use (int) (Math. Random () * 11).
- Declare a variable user Guess = 5.
- Print true if the random number matches the guess, otherwise print false.

Solution:

```

public class LAB_2 {
    public static void main(String[] args){
int Userguess =5;
        int Random_number=(int)((1+10-1)*Math.random());
        System.out.println("Random number bw 1 and 10 is "+Random_number);
        if(Random_number==Userguess){
            System.out.println("True");

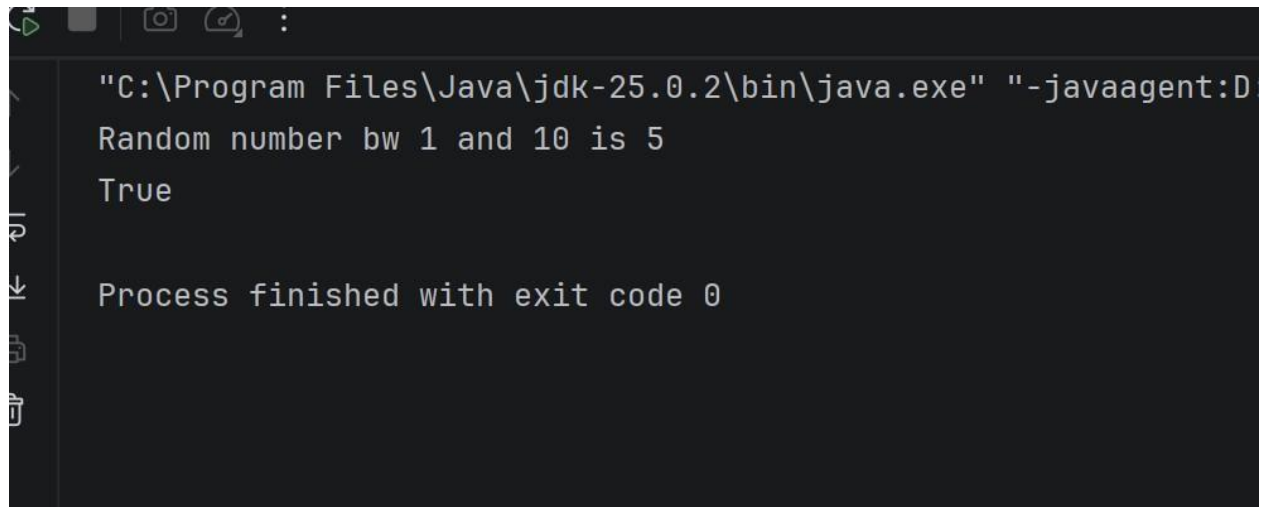
```

```

    }
else{
    System.out.println("False");
}
}
}

```

Output:



```

"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D
Random number bw 1 and 10 is 5
True

Process finished with exit code 0

```

Task No. 6:

Personalized Greeting (Strings & Concatenation)

Objective: Use different methods of string joining.

- Create variables for greeting, firstName, and lastName.
- Use the + operator to join them for one output.
- Use the .concat() method to join them for a second output.

Solution:

```

public class LAB_2{    public static
void main(String[] args){
    String Greetings="Assalam o Alaikum";
    String First_name="Muhammad";
    String Last_name="Rajab Raza";
    System.out.println(Greetings+" "+First_name+" "+Last_name);
    System.out.println(Greetings.concat(" "+First_name.concat("
"+Last_name))));

```

```
}  
}
```

Output:

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent  
Assalam o Alaikum Muhammad Rajab Raza  
Assalam o Alaikum Muhammad Rajab Raza  
  
Process finished with exit code 0
```

Task No. 7:

Time-of-Day Identifier (If...Else If) **Objective:**

Use conditional chains.

- Store an int variable hour (0 to 23).
- If hour is between 5 and 11, print "Good Morning".
- Else if hour is between 12 and 17, print "Good Afternoon".
- Else if hour is between 18 and 23, print "Good Evening".
- Use else to handle any other numbers as "Invalid Hour".

Solution:

```
public class LAB_2{  
    public static void main(String[] args) {  
        Scanner scan =new Scanner(System.in);  
        System.out.println("Enter an int variable hour (0 to 23):");  
        int hour= scan.nextInt();  
        System.out.println("The hour is " + hour);  
        if (hour <= 5 || hour >= 11) {  
            System.out.println("Good morning");  
        } else if (hour <= 12 || hour >= 17) {  
            System.out.println("Good afternoon");  
        } else if (hour <= 18 || hour >= 23) {
```



```

        System.out.println("Good evening");
    } else {
        System.out.println("Invalid Input");
    }
}
}

```

Output:

```

Enter an int variable hour (0 to 23):
5
The hour is 5
Good morning

Process finished with exit code 0

```

Task No. 8:

The Currency Switcher (Switch Case)

Objective: Practice multi-way branching.

- Create a variable String currency = "USD".
- Use a switch statement to print the full name of the currency:
 - ✦ "USD" -> "United States Dollar"
 - ✦ "GBP" -> "British Pound"
 - ✦ "EUR" -> "Euro"
 - ✦ Default -> "Unknown Currency"

Solution:

```

public class LAB_2{    public static
void main(String[] args){
    Scanner scan =new Scanner(System.in);
    System.out.println("Enter type of currency: ");
    String currency= scan.nextLine();
    switch(currency){        case "USD":
        System.out.println("United States Dollar");
    break;        case "GBP":
        System.out.println("British Pound");
    break;        case "EUR":

```



```

        System.out.println("Euro");
        break;
default:
    System.out.println("Unknown currency");
    }
    }
}

```

Output:

```

Enter type of currency:
USD
United States Dollar

Process finished with exit code 0
|

```

Task No. 9:

Comparison & Increment Lab (Operators)

Objective: Understand how values change.

- Declare int x = 10 and int y = 20.
- Use a comparison operator to print if x is not equal to y.
- Apply the increment operator ++ to x.
- Print the new value of x and check if it is now greater than 10.

Solution:

```

public class LAB_2{
    public static void main(String[] args){
int x=10,y=20;    if (x!=y){
        System.out.println(x);
    }
x++;
    System.out.println("Now the value of x is "+x);
    if(x>10){
        System.out.println("x is greater than 10");
    }
}

```

```
    }  
  }  
}  
  } }
```

Output:

```
10  
Now the value of x is 11  
x is greater than 10|  
  
Process finished with exit code 0
```

